

Deploying Django on Vercel (with PostgreSQL)

This guide explains how to deploy a Django project on Vercel including psycopg2 issues, PostgreSQL setup, environment variables, and migrations.

1. Requirements:

- Django project
- GitHub repo
- PostgreSQL database (Neon, Supabase, Prisma, etc.)
- Vercel account

2. psycopg2 issue:

Vercel cannot build psycopg2 because pg_config is missing.

Fix: use psycopg2-binary instead of psycopg2.

```
pip install psycopg2-binary
```

Remove psycopg2 from requirements.txt.

3. Django settings:

```
DATABASES = {  
    "default": dj_database_url.parse(os.environ["DATABASE_URL"])  
}
```

```
ALLOWED_HOSTS = [".vercel.app"]
```

```
CSRF_TRUSTED_ORIGINS = ["https://*.vercel.app"]
```

4. Static files:

Use whitenoise:

```
STATIC_ROOT = BASE_DIR / "staticfiles"
```

5. Vercel config:

Create vercel.json pointing to wsgi.py.

6. Environment variables (Vercel):

```
SECRET_KEY  
DEBUG=False  
DATABASE_URL  
ALLOWED_HOSTS
```

7. .env.production:

Store production secrets locally (DO NOT COMMIT).

8. Migrations:

Option A (recommended):

```
export $(cat .env.production | xargs)  
python manage.py migrate
```

Option B:

Use Vercel CLI

9. Deployment:

Push to GitHub → Import in Vercel → Deploy

10. Common errors:

- psycopg2 build error → use psycopg2-binary
- DisallowedHost → add .vercel.app
- CSRF error → add trusted origins
- Missing tables → run migrate